



Tensile properties of cooked meat sausages and their correlation with texture profile analysis (TPA) parameters and physico-chemical characteristics

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ABSTRACT

The possibilities of using breaking strength (BS) and energy to fracture (EF) for monitoring textural properties of some cooked meat sausages (chopped, mortadella and galantines) were studied. Texture profile analysis (TPA), folding test and physico-chemical measurements were also performed. Principal component analysis enabled these meat products to be grouped into three textural profiles which showed significant ($p < 0.05$) differences mainly for BS, hardness, adhesiveness and cohesiveness. Multivariate analysis indicated that BS, EF and TPA parameters were correlated ($p < 0.05$) for every individual meat product (chopped, mortadella and galantines) and all products together. On the basis of these results, TPA parameters could be used for constructing regression models to predict BS. The resulting regression model for all cooked meat products was $BS = -0.160 + 6.600 * \text{cohesiveness} - 1.255 * \text{adhesiveness} + 0.048 * \text{hardness} - 506.31 * \text{springiness}$ ($R^2 = 0.745$, $p < 0.00005$). Simple linear regression analysis showed significant coefficients of determination between BS ($R^2 = 0.586$, $p < 0.0001$) versus folding test grade (FG) and EF versus FG ($R^2 = 0.564$, $p < 0.0001$).

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1. Introduction

In recent years, consumers have demanded meat products that are safe, nutritious, convenient, rich in variety, attractive (in appearance, texture, odour and taste) and innovative. This stimulates interest in manufacturing cooked meat sausages by using new technologies and formulations, using different types of meat (pork, beef, poultry) and reducing levels of phosphate, salt and fat, all of which lead to beneficial effects on health (Desmond, 2006; Kemi, Karkkainen, & Lamberg-Allardt, 2006). These modifications in the manufacture of cooked meat sausages may affect the quality of the products (Farouk, Hall, Harrison, & Swan, 1999; Jiménez-Colmenero, 2000; Ruusunen & Puolanne, 2005), particularly texture (Jiménez-Colmenero, 2000).

Many instrumental methods have been developed to determine food textural properties (Bourne, 2002; Kilcast, 2004). Nowadays, the most commonly used instrumental method is, probably, the compression method of texture profile analysis (TPA), which mimics the conditions to which the material is subjected throughout the mastication process (Bourne, 1978; Scott-Blair, 1958).

The compression parameters obtained with TPA have been employed on cooked meat sausages by many authors as indices

to determine the quality of the finished product or to determine the textural property modifications due to new formulations (García, Cáceres, & Selgas, 2006; Kerr, Wang, & Choi, 2005; Mor-Mur & Yuste, 2003; Yilmaz, Simsek, & Isikli, 2002). However, the use of other texture instrumental methods could provide complementary valuable information about cooked meat sausages. For this, a textural instrumental method, the so-called tensile test, based on resistance of the sample to force deformation, has been developed (Bourne, 2002). Several tensile parameters can be obtained such as the maximum rupture force (maximum peak height resisted by the material), breaking strength (maximum rupture force by the cross-sectional area of the product) and energy to fracture (area under the deformation curve) (Bourne, 2002; Honikel, 1998).

The tensile test has been used to study the mechanical properties of whole meat, single muscle fibres and perimysial connective tissue (Christensen, Purslow, & Larsen, 2000; Christensen, Young, Lawson, Larsen, & Purslow, 2003; Lepetit & Culioli, 1994; Lewis & Purslow, 1989; Mutungi, Purslow, & Warkup, 1995; Willems & Purslow, 1996). Recently, the tensile test has been successfully used on meat products to obtain more textural property information on fermented sausages (Herrero et al., 2007) and meat spaghetti (Farouk, Zhang, & Waller, 2005). Tensile properties, such as breaking strength and energy to fracture, are important parameters of quality in meat sausages due to the increasing tendency of marketing previously sliced products. These slices can break easily during handling, processing and packing. If the

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breaking strength or the energy to fracture were less than the superficial adhesion force between the product and the surface of the processing equipment an important problem could arise because many of products could break leading to problems that can result in the processing line being stopped. Another undesirable example is when the force of adhesion of the product to packaging material or to another slice of the product is higher than the breaking strength or the energy to fracture because this results in distortion, disfigurement and breakage of the product, producing adverse reactions in consumers. These problems could occur in different meat products such as cooked meat sausages when they are sliced and vacuum packaged. However, no reports have been found about tensile measurements in these cooked meat products.

Therefore, the primary aim of this work was to apply tensile tests to cooked meat sausages to determine their tensile parameters (breaking strength and energy to fracture) and relating these results to those of TPA parameters, folding test score and physico-chemical characteristics, in order to provide a complete characterization of these products. Once the relationship between tensile and TPA parameters is established, multiple linear regression can be used to predict the tensile parameters using the TPA results as predictor variables. Thus, with a simple test (TPA) it should be possible to obtain data of tensile and compression textural parameters for cooked meat sausages.

2. Materials and methods

2.1. Description of the samples

Twenty-four samples (0.5 kg for each sample, vacuum packaged) of three different types of cooked meat sausages (chopped, mortadella and galantines) from different commercial brands were purchased in retail shops in Spain. The 24 samples were

bought three times (January, May and October) maintaining brands and meat sausage type. At every time of purchase, eight samples were of chopped (CH), nine of mortadella (MT), and seven of galantines (GL). Chopped is a heat-cured meat sausage manufactured with a mixture of ham chunks and trimmings and seasonings, ground together and then packaged into loaves. Mortadella is a finely hashed/ground heat-cured meat sausage which incorporates spices (including black pepper, myrtle berries, nutmeg and coriander) and sometimes small cubes of pork fat (principally the hard fat from the neck of the pig). A galantine is a French dish of boned stuffed meat, most commonly poultry or fish, that is poached and served cold, coated with aspic. Table 1 shows the meat origin, diameter and slice appearance of the different cooked meat sausages studied. The samples were kept at 5 °C until analysis. All the sausages were subjected to each type of physical, chemical and textural analysis.

2.2. Physico-chemical analysis

After removing the plastic case, chemical analyses were made in duplicate on all cooked sausages. About 200 g of sample were finely cut and some aliquots were used for the different analyses. The pH was determined in a distilled water homogenate (1:10) (w/v) of the sample (10 g) using a Crison Digit-501 pH meter (Crison Instruments LTD, Barcelona, Spain). Dry matter (DM) was determined by drying the sample at 110 °C to constant weight and the results are expressed as a percentage. Water activity (a_w) was measured using a Decagon CX1 hygrometer (Decagon Devices Inc., Pullman, WA, USA) at 25 °C. The total fat content of the samples was determined by cold extraction in chloroform and methanol in the presence of antioxidant BHT as described by Hanson and Olley (1963) and was quantified gravimetrically. Results are expressed as percentage of dry matter (DM).

Table 1
Characteristics of the cooked meat sausages analysed

Sample ^a	Product	Meat origin	Diameter (cm)	Slice appearance
CH1	Chopped	Pork	5.5 ± 0.2	~60% fine emulsion with coarse meat (0.5–2 cm ²)
CH2	Chopped	Turkey	10.0 ± 0.4	
CH3	Chopped	Beef	5.0 ± 0.3	
CH4	Chopped	Iberian pork	5.0 ± 0.2	~70% fine emulsion with flat strip meat (0.5 × 1.5 cm) and coarse fat
CH5	Chopped	Pork	6.0 ± 0.2	~55% fine emulsion with coarse meat (1–1.5 cm ²)
CH6	Chopped	Pork	10.5 ± 0.5	~80% fine emulsion with coarse meat (0.5–2 cm ²)
CH7	Chopped	Iberian pork	10.0 ± 0.5	~60% fine emulsion with irregular coarse meat (0.3–2 cm ²)
CH8	Chopped	Pork	10.0 ± 0.4	~40–50% fine emulsion with cubes of meat and fat (~1 cm ²)
MT1	Mortadella	Pork	10.0 ± 0.3	~80% fine emulsion with green-olives
MT2	Mortadella	Iberian pork	6.0 ± 0.2	~70% fine emulsion with cubes of fat (~1 cm ²)
MT3	Mortadella	Pork and beef (traditional type)	5.3 ± 0.2	
MT4	Mortadella	Pork and beef (traditional type)	5.5 ± 0.3	
MT5	Mortadella	Iberian pig	5.7 ± 0.2	~60–65% fine emulsion with coarse meat (0.1–0.3 cm ²)
MT6	Mortadella	Turkey	5.8 ± 0.2	~65–70% fine emulsion with coarse meat (~0.1–0.3 cm ²)
MT7	Mortadella	Pork	16.0 ± 0.5	~65–70% fine emulsion with cubes of fat (~0.25 cm ²)
MT8	Mortadella	Turkey	11.5 ± 0.4	~65% fine emulsion and coarse meat (0.01–0.1 cm ²). Foamy aspect and many small cavities
MT9	Mortadella	Turkey	12.5 ± 0.3	~60% fine emulsion with coarse meat (~0.05–0.5 cm ²)
GL1	Galantine	Pork and fish	11.5 ± 0.4	~80% fine emulsion with kamaboko (cube pieces ~2.25 cm ²)
GL2	Galantine	Chicken	17.5 ± 0.4	~10–15% fine emulsion with coarse meat (1–2 cm ²) and pistachio nut
GL3	Galantine	Chicken	11.0 ± 0.4	~10% fine emulsion with irregular meat portions (~1–12 cm ²)
GL4	Galantine	Duck	12.0 ± 0.5	~60–70% fine emulsion with irregular coarse meat (~4–12 cm ²)
GL5	Galantine	Chicken	15.0 ± 0.5	~65% fine emulsion with irregular coarse meat (~0.5–8 cm ²) with fine/coarse fat and herbs
GL6	Galantine	Pork and fish	11.0 ± 0.4	~30% fine emulsion with irregular kamaboko portions (~1–16 cm ²) and fine/coarse red pepper
GL7	Galantine	Chicken	12.0 ± 0.3	~60% fine emulsion with irregular coarse meat (~0.5–4 cm ²)

^a CH = chopped, MT = mortadella, GL = galantine.

2.3. Textural analysis

Texture profile analysis (TPA), tensile test and folding tests were carried out at about 22 °C. All textural procedures involved discarding the external plastic case of the cooked sausages.

TPA and tensile test were performed using a TA.XT2i SMS Stable Micro Systems Texture Analyser (Stable Microsystems Ltd., Surrey, England) with the *Texture Expert* programmes.

2.3.1. Texture profile analysis (TPA)

In general, four cylinders 1.5 cm high and 2 cm wide were prepared from every sample. A double compression cycle test was performed up to 50% compression of the original portion height with an aluminium cylinder probe of 2 cm diameter. A time of 5 s was allowed to elapse between the two compression cycles. Force–time deformation curves were obtained with a 25 kg load cell applied at a cross-head speed of 2 mm/s. The following parameters were quantified (Bourne, 1978): hardness (N) maximum force required to compress the sample, springiness (m), ability of the sample to recover its original form after deforming force was removed, adhesiveness (N s), area under the abscissa after the first compression, and cohesiveness, extent to which the sample could be deformed prior to rupture.

2.3.2. Tensile test

In general, five pieces were cut in a dumbbell shape, approximately 7.5 cm long, 2 cm wide in the narrowest zone and 0.2 cm thickness per sample. A load cell of 5 kg was employed. For analysis, one tensile grip (A/TGT) was fixed to the base of the textural analyser, while the other one was attached to the load cell. Initial grip separation was 12.5 mm and cross-head speed was 1.0 mm/s until rupture (Herrero et al., 2007). Each sample was placed between both tensile grips on the textural analyzer. Rupture force was taken as the maximum force peak height (N) required for breaking the sample. Breaking strength (N/cm²) was obtained dividing the rupture force by the cross-sectional area (thickness × width) of the portions. Energy to fracture (N mm) was calculated as the area under the deformation curve (Honikel, 1998).

2.3.3. Folding test

This test was conducted by folding a 3 mm thick slice of meat sausage slowly in half, and then in half again to examine the structural failure of the sample. The evaluation was performed in accordance with a five-point grade system (Suzuki, 1981) as follows: grade (5), no crack when folded into quadrants; grade (4), no crack when folded in half; grade (3), crack develops gradually when folded in half; grade (2), crack develops immediately when folded in half; grade (1), crumbles when pressed by finger.

2.4. Statistical analysis

An individual cooked meat sausage was the experimental unit for analysis of all data. To check the normal distribution (90% confidence) of samples, the Shapiro–Wilks test was applied. When samples fitted the normal distribution, one-way ANOVA analysis was performed. When samples did not fit the normal distribution, the Kruskal–Wallis test was used to test the null hypothesis that the medians of variables within each of the levels of samples were the same. Duncan's test to multiple mean comparisons (at 95% or 99% of confidence level), Pearson product moment correlations, principal component, simple and multiple regression analysis (using a Durbin–Watson statistic tests, at 95% of confidence level) were performed to determine the relationships between data obtained by tensile test, TPA and physico-chemical analysis. The statistical analysis was carried out using a Statgraphics Plus version 5.0. The analyses were conducted across all sausages types. Data

were presented as the mean of each sample and the standard deviation (SD) of the mean.

3. Results and discussion

3.1. Physico-chemical analysis

Dry matter, fat content (% dry matter), water activity (a_w) and pH values of the different cooked sausages are shown in Table 2. Significant differences ($p < 0.01$) in all these physico-chemical parameters were found (Table 2). These differences ($p < 0.01$) could be attributed to variations in the product formulation (Table 1) and probably are not due to the type of cooked meat sausage. Results show that dry matter ranged from 28.1% to 49.1% wet matter and fat contents varied from 19.2% to 50.3% dry matter (Table 2). In general, galantines had low dry matter and fat values (Table 2) with values close to that reported by others authors (Mielnik, Aaby, Rolfsen; Ellekjær, & Nilsson, 2002; Yilmaz et al., 2002). The cooked sausages analysed (Table 1) could be grouped according to fat content as: low fat (<25%), medium fat (25–35%) and high fat (>35%). According to this criterion, only 8.3% of the samples belonged to the low fat group, 33.3% to the medium fat category, and 58.4% to the high fat group. Samples of chopped and galantines were distributed in all three groups, although, in general, chopped samples showed higher fat content than galantines. It was also observed that almost all mortadella samples were classified as high fat.

The water activity and pH values of the commercial cooked products studied ranged from 0.946 to 0.986 and 6.58 to 7.05, respectively.

3.2. Textural analysis

Textural properties of the cooked meat sausages are shown in Table 3. Results from the tensile and TPA analysis showed significant variations ($p < 0.05$) indicating a great dispersion of textural properties between all samples studied. The breaking strength (BS) and energy to fracture (EF) fell between 0.03 and 4.67 N/cm²

Table 2

Dry matter (DM, % wet matter), fat content (% DM), water activity (a_w) and pH of cooked meat sausages

Samples ^A	DM	Fat content	a_w	pH
CH1	49.1 ± 0.5a	41.6 ± 1.6a,b	0.960 ± 0.001c,d	6.62 ± 0.14c
CH2	28.1 ± 0.5e	19.2 ± 1.4e	0.978 ± 0.001a,b	6.65 ± 0.14b,c
CH3	34.1 ± 2.2c,d	34.8 ± 4.0c	0.974 ± 0.001a,b	6.63 ± 0.10c
CH4	45.2 ± 0.7a,b	46.8 ± 1.1a	0.966 ± 0.001c	6.81 ± 0.14b,c
CH5	35.9 ± 1.4c	34.8 ± 0.5c	0.975 ± 0.001a,b	6.63 ± 0.02c
CH6	33.9 ± 1.4c,d	42.2 ± 3.0a,b	0.966 ± 0.001c	6.75 ± 0.21b,c
CH7	42.3 ± 0.4b	50.3 ± 1.9a	0.959 ± 0.003c,d	6.88 ± 0.04a
CH8	33.2 ± 1.3c,d	36.8 ± 3.3b	0.971 ± 0.002 b	6.72 ± 0.02b,c
MT1	33.7 ± 0.1c,d	43.8 ± 1.6a	0.969 ± 0.001b,c	6.65 ± 0.02b,c
MT2	39.8 ± 0.1b,c	45.8 ± 0.5a	0.980 ± 0.001a	6.75 ± 0.03b,c
MT3	37.1 ± 0.7c	41.6 ± 0.6a,b	0.970 ± 0.001b	6.76 ± 0.05b,c
MT4	37.2 ± 0.8c	49.2 ± 0.5a	0.970 ± 0.001b	6.76 ± 0.05b,c
MT5	31.1 ± 1.6d	44.4 ± 5.1a	0.974 ± 0.001a,b	7.05 ± 0.04a
MT6	27.8 ± 0.1e	35.3 ± 2.1c	0.977 ± 0.001a,b	6.81 ± 0.03b,c
MT7	47.2 ± 3.2a	46.9 ± 1.0a	0.946 ± 0.003d	6.61 ± 0.05c
MT8	31.3 ± 0.2d	38.9 ± 1.6b	0.964 ± 0.001c	6.72 ± 0.06b,c
MT9	30.9 ± 0.8d	32.7 ± 0.1c	0.968 ± 0.001b,c	6.82 ± 0.08a,b
GL1	36.9 ± 0.1c	35.8 ± 0.5c	0.961 ± 0.001c,d	6.83 ± 0.01a,b
GL2	24.6 ± 1.7f	20.6 ± 1.1e	0.967 ± 0.001c	6.62 ± 0.11c
GL3	32.9 ± 1.1c,d	32.0 ± 0.9c	0.966 ± 0.003c	6.69 ± 0.01b,c
GL4	42.2 ± 0.3b	32.1 ± 0.2c	0.966 ± 0.001c	6.58 ± 0.07d
GL5	35.1 ± 0.8c,d	39.5 ± 0.8b	0.963 ± 0.001c,d	6.83 ± 0.01a,b
GL6	31.0 ± 0.8d	29.2 ± 2.0d	0.966 ± 0.001c	6.75 ± 0.03b,c
GL7	32.0 ± 1.2c,d	23.7 ± 1.3d	0.986 ± 0.001a	6.81 ± 0.14b,c

Different letters in the same column indicate significant differences ($p < 0.001$).

^A ACH, chopped; MT mortadella; GL, galantines.

Table 3

Textural properties of cooked meat sausages

Samples ^A	Breaking strength (N cm ⁻²)	Energy to fracture (N mm)	Hardness (N)	Adhesiveness (N s)	Cohesiveness	Springiness 10 ⁻² m	Foldingtest grade
CH1	0.24 ± 0.08d	2.17 ± 0.47c	61.2 ± 3.6c,d	-0.02 ± 0.02a	0.21 ± 0.06d	0.55 ± 0.06b,c	3
CH2	0.03 ± 0.01d	0.68 ± 0.08e	23.7 ± 1.7a	-0.17 ± 0.06b,c	0.26 ± 0.07c	0.47 ± 0.09c,d	3
CH3	1.70 ± 0.26b	5.14 ± 0.98b	50.8 ± 4.9d,e	-0.06 ± 0.01a	0.28 ± 0.04c	0.52 ± 0.07c	3
CH4	1.87 ± 0.38a,b	5.17 ± 1.54b	65.1 ± 6.3c	-0.53 ± 0.06d,e	0.34 ± 0.09b	0.54 ± 0.06b,c	5
CH5	2.17 ± 0.20a,b	6.22 ± 0.98a,b	44.0 ± 3.7e	-0.03 ± 0.02a	0.25 ± 0.03c	0.45 ± 0.05d	3
CH6	1.92 ± 0.36a,b	5.55 ± 1.54b	42.0 ± 5.6e	-0.04 ± 0.01a	0.27 ± 0.09c	0.54 ± 0.10b,c	3
CH7	3.50 ± 0.50a	18.35 ± 2.06a	70.1 ± 5.8b	-0.95 ± 0.10f	0.31 ± 0.07c	0.58 ± 0.07b	5
CH8	2.80 ± 0.18a,b	7.90 ± 1.12a,b	54.6 ± 2.4d	-1.08 ± 0.02f	0.32 ± 0.04b	0.61 ± 0.02a,b	5
MT1	1.64 ± 0.75b	4.36 ± 1.37b	41.9 ± 3.4e	-0.37 ± 0.08d,e	0.40 ± 0.06a,b	0.48 ± 0.03c	4
MT2	0.15 ± 0.02d	1.04 ± 0.23d	37.7 ± 5.2e,f	-0.24 ± 0.02b,c	0.18 ± 0.09d	0.46 ± 0.02c,d	3
MT3	0.96 ± 0.44c	2.28 ± 0.35c	40.6 ± 2.6e	-0.18 ± 0.03b,c	0.39 ± 0.07a,b	0.62 ± 0.05a,b	4
MT4	0.84 ± 0.27c	2.13 ± 0.32c	41.7 ± 2.6e	-0.20 ± 0.03b,c	0.40 ± 0.07a,b	0.62 ± 0.05a,b	4
MT5	4.25 ± 0.40a	16.09 ± 1.88a	55.3 ± 4.2d	-0.60 ± 0.03e	0.45 ± 0.03a	0.67 ± 0.06a	5
MT6	3.36 ± 0.46a	14.64 ± 1.81a	41.9 ± 2.9e	-0.40 ± 0.02d,e	0.50 ± 0.02a	0.67 ± 0.05a	5
MT7	3.59 ± 0.51a	11.44 ± 2.31a,b	78.2 ± 2.4a	-0.31 ± 0.09c,d	0.48 ± 0.04a	0.64 ± 0.04a,b	5
MT8	2.03 ± 0.22a,b	6.86 ± 2.19a,b	45.2 ± 3.5e	-0.24 ± 0.04b,c	0.45 ± 0.03a	0.53 ± 0.03c	5
MT9	2.73 ± 0.19a,b	10.05 ± 1.89a,b	40.6 ± 2.5e	-0.85 ± 0.03f	0.39 ± 0.08a,b	0.58 ± 0.03b	5
GL1	4.67 ± 0.24a	15.36 ± 1.31a	54.5 ± 3.6d	-0.60 ± 0.10e	0.40 ± 0.08a,b	0.62 ± 0.07a,b	5
GL2	1.65 ± 0.49b	4.25 ± 1.71b	32.5 ± 4.7f	-0.03 ± 0.01a	0.34 ± 0.05b	0.50 ± 0.07c	3
GL3	1.10 ± 0.27b	4.86 ± 1.25b	34.9 ± 2.0f	-0.03 ± 0.01a	0.28 ± 0.07c	0.46 ± 0.02d	3
GL4	0.57 ± 0.28c	2.24 ± 0.87c	20.9 ± 1.9a	-0.24 ± 0.10b,c	0.27 ± 0.03c	0.48 ± 0.05c,d	3
GL5	0.96 ± 0.18c	1.88 ± 0.27c	21.0 ± 2.1a	-0.13 ± 0.06a,b	0.37 ± 0.09b	0.64 ± 0.06a,b	3
GL6	1.24 ± 0.47b	4.30 ± 1.08b	37.4 ± 2.2e,f	-0.59 ± 0.08e	0.34 ± 0.09b	0.61 ± 0.06a,b	4
GL7	0.08 ± 0.03d	0.72 ± 0.35e	50.8 ± 7.2d,e	-0.10 ± 0.03a,b	0.46 ± 0.04a	0.68 ± 0.03a	3

Different letters in the same column indicate significant differences ($p < 0.05$).^A CH, chopped; MT, mortadella; GL, galantines.

and 0.68 and 18.35 N mm, respectively. The most representative values of BS and EF were those in the range of 1–3 N/cm² and 4–12 N mm, respectively. About 50% of cooked meat sausages showed BS and EF values within these intervals. The galantines had low-values of BS and EF, except for the GL1 sample. Hardness ranged from 20.9 to 78.2 N, with around 63% of the samples showing values lower than 50 N (Table 3). As in the case of BS and EF, the galantines had low hardness, except for the GL1 sample. Values for adhesiveness values ranged from -0.02 to -1.08 N mm, indicating a great variation in this textural property amongst all samples studied. The cohesiveness ranged from 0.21 to 0.34, from 0.18 to 0.50 and from 0.27 to 0.46 for chopped, mortadella and galantines, respectively. Springiness values showed less variation with 88% of samples ranging from 0.45 to 0.65 10⁻² m. The range of TPA values shown in Table 3 are similar to that reported by some authors for different cooked sausages (García et al., 2006; Kerr et al., 2005; Mor-Mur & Yuste, 2003; Yılmaz et al., 2002). According to the folding test (Table 3) samples were assigned to grades 3, 4 and 5. Only 33% of the products did not crack when folded into quadrants and fell into the maximum grade (5). However, 62.5% of chopped and about 71% of the galantines scored grade 3 (crack develops gradually when folded in half), while only 11% of the mortadella scored this grade. The majority of the mortadella, 55.5%, was in grade 5.

Pearson product moment correlations among the texture variables shown in Table 3 were performed. Results indicated that BS, hardness, cohesiveness and adhesiveness were strongly correlated ($p < 0.0001$). After applying an analysis of principal components using the data obtained for these textural properties of cooked sausages as criterion of association, it was possible to distinguish three different clusters. The mean values of BS, hardness, cohesiveness and adhesiveness of the different sausages included in each cluster and the mean values of springiness were calculated and then plotted. As it can be observed in Fig. 1, three different ($p < 0.05$) textural profiles, arbitrary named 1, 2, and 3, were obtained. These textural profiles showed significant ($p < 0.05$) differences for textural properties, mainly for BS, hardness, adhesiveness and cohesiveness (Fig. 1). The main textural differences ($p < 0.05$)

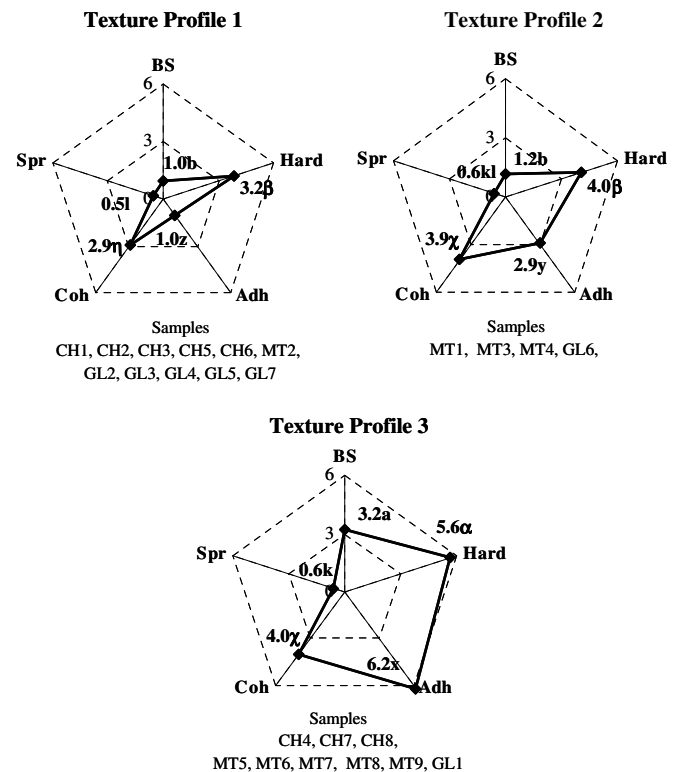


Fig. 1. Textural profiles (1, 2 and 3) and mean values of different textural parameters from cooked meat sausages (CH, chopped; MT, mortadella; GL, galantines). BS: Breaking strength (N cm⁻²), Hard: Hardness (10⁻¹ N), Adh: Adhesiveness (-10 N s), Coh: Cohesiveness (×10), Spr: Springiness (10⁻² m). Mean values with different letter differ significantly ($p < 0.05$): a, b for BS; α, β for Hard; x, y, z for Adh; χ, η for Coh; k, l for Spr.

were observed between the textural profiles 1 and 3. Textural profile 1 showed lower values ($p < 0.05$) of BS, hardness, adhesiveness, cohesiveness and springiness than textural profile 3. Profile 2

showed intermediate textural behaviour between profiles 1 and 3 with similar ($p > 0.05$) values of BS and hardness to textural profile 1, and similar values ($p > 0.05$) of cohesiveness and springiness to profile 3, and intermediate values of adhesiveness (Fig. 1).

About 46% of cooked meat sausages were included in textural profile 1 and only a 17% of the samples analysed were grouped in the textural profile 2 (Fig. 1). It could be observed that mortadella and galantine samples were very heterogeneous products because they are included in the three textural profiles although the majority of mortadella (67%) belonged to profile 3 and 71% of galantines were included in profile 1 (Fig. 1). Chopped was a more homogeneous product with samples belonging to textural profiles 1 (62.5%) and 3 (Fig. 1).

The folding test results showed that samples included in textural profile 1 scored grade 3 (crack develops gradually when folded in half). These results could be associated with low-values of BS and TPA parameters of the textural profile 1. All products belonging to profile 1 (Fig. 1) were visually characterized (Table 1) by a matrix of a fine emulsion that included large meat pieces and other materials. Samples classified in the textural profile 3 (Fig. 1) scored grade 5 (Table 3) in the folding test (no crack when folded into quadrants), which is the maximum grade, indicating good gelling ability. These scores in the folding test could be explained by the high values of breaking strength and TPA parameters of the textural profile 3. The visual and rheological analysis of profile 3 products indicated that its textural behaviour could be associated with a strong matrix of fine emulsion with or without coarse meat (Table 1). Folding test results indicated that samples of textural profile 2 scored grade 4, no crack when folded in half. These textural profiles had intermediate values for BS and TPA parameters. The visual (Table 1) and the rheological analysis (Table 3, Fig. 1) could indicate that texture behav-

iour of samples of profile 2 are associated with a well gelled matrix which included materials of different origin and size (kamaboko, olives, etc.) or fat cubes which are easily liberated during the folding test.

3.3. Linear regression analysis

The multiple linear regression analyses (Table 4), using the different textural parameters as dependent variables and values of dry matter, fat contents and a_w as independent variables, revealed a significant relation between hardness ($R^2 = 0.308$, $p < 0.05$), cohesiveness ($R^2 = 0.440$, $p < 0.0005$), breaking strength ($R^2 = 0.330$, $p < 0.005$), energy to fracture ($R^2 = 0.234$, $p < 0.05$) and folding test grade ($R^2 = 0.334$, $p < 0.05$), versus DM, fat content and a_w (Table 4). The statistical significance of the t -values indicates that DM and fat content participate in all textural parameters previously mentioned. These results are in agreement with some authors who have described the relationship between fat content and textural properties of cooked meat sausages (Giese, 1996; Jiménez-Colmenero, 2000; Rust & Olson 1988). Results of the t -values showed a significant correlation between a_w and cohesiveness, BS and EF. In previous work (Herrero et al., 2007) it was found that in dry fermented sausages, a_w is highly correlated with breaking strength, while dry matter is correlated with cohesiveness, springiness and adhesiveness.

Simple linear regression analysis was performed to determine the degree of association between BS and EF versus folding test grade (FG). Significant ($p < 0.0001$) coefficients of determination between tensile and folding test parameters were obtained ($R^2 = 0.586$ for BS versus FG and $R^2 = 0.564$ for EF versus FG). The equations of the fitted models were $BS = -2.533 + 1.120 \cdot FG$ (correlation coefficient, $R = 0.765$, R^2 adjusted for degrees of

Table 4

Multiple linear regression analysis of textural parameters versus dry matter (DM, % wet matter), fat content (% DM) and water activity (a_w) of cooked meat sausages

Dependent variable	R^2	SE	Independent variable	Regression coefficient	t -Values	β -Values
Hardness	0.308*	8.917	Constant	22.496		
			DM	-0.929	-2.762*	-0.432
			Fat content	0.766	3.148**	0.518
			a_w	21.717	-1.026	0.012
Cohesiveness	0.440***	0.069	Constant	7.396		
			DM	-0.012	-4.184***	-0.637
			Fat content	0.007	3.896***	0.561
			a_w	-7.119	-3.639**	-0.547
Adhesiveness	0.080	0.382	Constant	3.281		
			DM	0.001	0.054	0.009
			Fat content	-0.014	-1.080	-0.225
			a_w	-3.305	-0.268	-0.057
Springiness	0.115	0.001	Constant	0.025		
			DM	5.011×10^{-5}	-0.942	0.020
			Fat content	6.554×10^{-5}	1.954	0.039
			a_w	-0.020	-0.587	-0.114
Breaking strength	0.330**	1.443	Constant	106.383		
			DM	-0.119	-2.163*	-0.330
			Fat content	0.097	2.785*	0.422
			a_w	-107.14	-2.977**	0.438
Energy to fracture	0.234*	5.019	Constant	303.271		
			DM	-0.467	-2.332*	-0.362
			Fat content	0.348	2.733*	0.414
			a_w	-302.775	-2.222*	-0.350
Folding test grade	0.334*	0.81	Constant	44.16		
			DM	-0.075	-1.943*	-0.003
			Fat content	0.06	2.491*	0.002
			a_w	-41.28	-1.67	-0.152

$n = 72$; SE = standard error; R^2 = coefficient of determination (correlation coefficient square).

* $p < 0.05$.

** $p < 0.005$.

*** $p < 0.0005$.

freedom = 0.567, mean absolute error = 0.751) and EF = $-10.182 + 4.235 \cdot \text{FG}$ ($R = 0.751$, R^2 adjusted for degrees of freedom = 0.545, mean absolute error = 2.85). Table 5 shows the multiple linear regression analysis of the BS and EF versus TPA parameters (cohesiveness, adhesiveness, hardness and springiness) for each cooked meat sausage type (chopped, mortadella or galantines) and for all samples. In the chopped samples, a significant multiple linear regression model ($R^2 = 0.733$, $p < 0.00005$) was found between breaking strength and TPA parameters, while energy to fracture was not significantly correlated ($p > 0.05$) with parameters obtained by TPA analysis (Table 5). Student's t -values of cohesiveness, adhesiveness, hardness and springiness partial regression coefficients were significant [BS versus cohesiveness ($p < 0.0005$), BS versus adhesiveness ($p < 0.0005$), BS versus hardness ($p < 0.00005$) and BS versus springiness ($p < 0.00005$)]. However, β values suggest

that cohesiveness, hardness and springiness values have direct influence on the BS determination.

A highly significant multiple linear regression was found between breaking strength ($R^2 = 0.864$, $p < 0.00005$) and energy to fracture ($R^2 = 0.809$, $p < 0.00005$) and TPA parameters in mortadella samples (Table 5). In these meat products, Student's t -values of cohesiveness ($p < 0.005$), adhesiveness ($p < 0.005$) and hardness ($p < 0.00005$) partial regression coefficients were significant versus BS, while all TPA parameters [(cohesiveness ($p < 0.0005$), adhesiveness ($p < 0.00005$), hardness ($p < 0.05$), and springiness ($p < 0.05$)] partial regression coefficients were significant versus EF. In addition, β values suggest that hardness and adhesiveness are the most important TPA parameters in BS and EF determination, respectively.

In the galantine samples, it was possible to find a high significant multiple linear regression ($R^2 = 0.937$, $p < 0.05$) between

Table 5

Multiple linear regression analysis of tension mechanical parameters (breaking strength and energy to fracture) versus texture profile analysis (TPA) parameters of cooked meat sausages

Dependent variable	R^2	SE	Independent variable	Regression coefficient	t -Values	β - Values
<i>Chopped</i>						
Breaking strength	0.733 ^{****}	0.651	Constant	−0.148		
			Cohesiveness	6.455	4.065 ^{***}	0.694
			Adhesiveness	−3.285	−3.360 ^{***}	−0.706
			Hardness	0.048	5.735 ^{****}	0.866
			Springiness	−506.07	−5.196 ^{****}	0.999
Energy to fracture	0.573	4.796	Constant	−9.109		
			Cohesiveness	6.307	0.407	0.092
			Adhesiveness	−2.681	−0.584	−0.205
			Hardness	0.258	1.710 ^{***}	0.464
			Springiness	44.884	0.058	0.030
<i>Mortadella</i>						
Breaking strength	0.864 ^{****}	0.535	Constant	−3.817		
			Cohesiveness	6.359	3.003 ^{**}	0.503
			Adhesiveness	−2.890	−3.798 ^{**}	0.574
			Hardness	0.061	6.510 ^{****}	0.999
			Springiness	−97.42	0.498	−0.147
Energy to fracture	0.809 ^{****}	2.638	Constant	−24.028		
			Cohesiveness	33.289	3.914 ^{***}	0.898
			Adhesiveness	−19.206	−5.066 ^{****}	0.998
			Hardness	0.106	2.135 [*]	0.440
			Springiness	1038.82	1.937 [*]	0.590
<i>Galantines</i>						
Breaking strength	0.937 [*]	0.206	Constant	2.930		
			Cohesiveness	−3.068	−1.708	−0.650
			Adhesiveness	−1.218	−3.879 ^{**}	−0.889
			Hardness	−0.012	−1.434	−0.583
			Springiness	−86.20	−0.969	−0.436
Energy to fracture	0.509	2.973	Constant	10.395		
			Cohesiveness	−11.530	−0.451	−0.198
			Adhesiveness	1.097	0.251	0.111
			Hardness	0.036	0.333	0.148
			Springiness	−562.72	−0.507	−0.222
<i>Combination of all cooked meat sausages</i>						
Breaking strength	0.745 ^{****}	0.643	Constant	−0.160		
			Cohesiveness	6.600	4.462 ^{***}	0.633
			Adhesiveness	−1.255	−3.670 ^{***}	−0.509
			Hardness	0.048	6.118 ^{****}	0.834
			Springiness	−506.31	−5.271 ^{****}	−0.778
Energy to fracture	0.491 ^{****}	3.921	Constant	−4.487		
			Cohesiveness	−11.530	1.797 [*]	0.264
			Adhesiveness	1.097	−2.443 [*]	−0.355
			Hardness	0.036	3.861 ^{***}	0.531
			Springiness	−562.72	−1.680	−0.251

SE = standard error; R^2 = Coefficient of determination (Correlation coefficient square).

* $p < 0.05$.

** $p < 0.005$.

*** $p < 0.0005$.

**** $p < 0.00005$.

breaking strength and TPA parameters, although only Student's *t*-value of adhesiveness partial regression coefficient was significant versus BS. Energy to fracture was not significantly correlated ($p > 0.05$) with TPA parameters (Table 5) for galantines.

For all cooked meat sausages, a highly significant multiple linear regression was found between breaking strength ($R^2 = 0.745$, $p < 0.00005$) and TPA parameters (Table 5). The Student's *t*-values of cohesiveness, adhesiveness, hardness and springiness partial coefficients were significant [BS versus cohesiveness ($p < 0.0005$), BS versus adhesiveness ($p < 0.0005$), BS versus hardness ($p < 0.00005$), BS versus springiness ($p < 0.00005$)]. In addition, β values suggest that hardness plays the most important role in breaking strength determination. Also, a significant regression was found between energy to fracture ($R^2 = 0.491$, $p < 0.00005$) and TPA parameters (Table 5) but only Student's *t*-values of adhesiveness and hardness partial coefficients were significant [EF versus adhesiveness ($p < 0.05$) and EF versus hardness ($p < 0.0005$)].

Therefore, the best regression model to predict tensile properties for cooked meat sausages is using BS as the dependent variable and TPA parameters as independent variables. The resulting regression model is $BS = -0.160 + 6.600 \times \text{cohesiveness} - 1.255 \times \text{adhesiveness} + 0.048 \times \text{hardness} - 506.31 \times \text{springiness}$. The correlation coefficient *R* was 0.863, the R^2 adjusted for degrees of freedom 0.715, and the mean absolute error was 0.511. Results of the multivariate analysis confirm that TPA parameters chosen were relevant for constructing regression models to predict BS for cooked meat sausages. Therefore with only a TPA analysis it could be possible to obtain both the TPA and tensile parameters such as the breaking strength.

4. Conclusions

The determination of breaking strength (BS) and the energy to fracture (EF) by tensile test can be used together with the TPA, to determine textural properties of cooked meat sausages. With these analyses complementary information is obtained, which permits grouping of cooked meat sausages into three different textural profiles. These textural profiles are characterized by the values BS, hardness, adhesiveness and cohesiveness.

The multivariate analysis confirms that TPA parameters (cohesiveness, adhesiveness, hardness and springiness) could be used to construct regression models to predict breaking strength. Therefore, with only a TPA analysis it could be possible to obtain both the TPA and tensile parameters such as the breaking strength.

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References

- Bourne, M. C. (1978). Texture profile analysis. *Food Technology*, 32, 62–66.
- Bourne, M.C. (2002). Principles of objective texture measurement. In M. C. Bourne (Ed.), *Food texture and viscosity: Concept and measurement* (pp. 107–188). San Diego, USA.
- Christensen, M., Purslow, P. P., & Larsen, L. M. (2000). The effect of cooking temperature on mechanical properties of whole meat, single muscle fibres and perimysial connective tissue. *Meat Science*, 55, 301–307.
- Christensen, M., Young, R. D., Lawson, M. A., Larsen, L. M., & Purslow, P. P. (2003). Effect of added (μ -calpain and post-mortem storage on the mechanical properties of bovine single muscle fibres extended to fracture. *Meat Science*, 66, 105–112.
- Desmond, E. (2006). Reducing salt: A challenge for the meat industry. *Meat Science*, 74, 188–196.
- Farouk, M. M., Hall, W. K., Harrison, M., & Swan, J. E. (1999). Instrumental and sensory measurement of beef patty and sausage texture. *Journal of Muscle Foods*, 10, 17–28.
- Farouk, M. M., Zhang, S. X., & Waller, J. (2005). Meat spaghetti tensile strength and extensibility as indicators of the manufacturing quality of thawed beef. *Journal of Food Quality*, 28, 452–466.
- García, M. L., Cáceres, E., & Selgas, M. D. (2006). Effect of inulin on the textural and sensory properties of mortadella, a Spanish cooked meat product. *International Journal of Food Science and Technology*, 41, 1207–1215.
- Giese, J. (1996). Fats, oils and fat replacers. *Food Technology*, 50, 78–83.
- Hanson, S. W. F., & Olley, J. (1963). Application of the Bligh and Dyer method of lipid extraction to tissue homogenates. *Biochemical Journal*, 89, 101–120.
- Herrero, A. M., Ordóñez, J. A., Romero de Ávila, M. D., Herranz, B., de la Hoz, L., & Cambero, M. I. (2007). Breaking strength of dry fermented sausages and their correlation with Texture Profile Analysis (TPA) and physico-chemical characteristics. *Meat Science*, 77, 331–338.
- Honikel, K. O. (1998). Reference methods for the assessment of physical characteristics of meat. *Meat Science*, 49, 447–457.
- Jiménez-Colmenero, F. (2000). Relevant factors in strategies for fat reduction in meat products. *Trends in Food Science and Technology*, 11, 56–66.
- Kemi, V. E., Karkkainen, M. U., & Lamberg-Allardt, C. J. E. (2006). High phosphorus intakes acutely and negatively affect Ca^{2+} and bone metabolism in a dose-dependent manner in healthy young females. *British Journal of Nutrition*, 96, 545–552.
- Kerr, W. L., Wang, X., & Choi, S. G. (2005). Physical and sensory characteristics of low-fat Italian sausage prepared with hydrated oat. *Journal of Food Quality*, 28, 62–77.
- Kilcast, D. (2004). Force/deformation techniques for measuring texture. In D. Kilcast (Ed.), *Texture in food* (Vol. 2, pp. 109–145). Abington, Cambridge UK: Woodhead Publishing Ltd.
- Lepetit, J., & Culoli, J. (1994). Mechanical properties of meat. *Meat Science*, 36, 203–237.
- Lewis, G. J., & Purslow, P. P. (1989). The strength and stiffness of perimysial connective tissue isolated from cooked beef muscle. *Meat Science*, 26, 255–269.
- Mielnik, M. B., Aaby, K., Rolfsen, K., Ellekjær, M. R., & Nilsson, A. (2002). Quality of comminuted sausages formulated from mechanically deboned poultry meat. *Meat Science*, 61, 73–84.
- Mor-Mur, M., & Yuste, J. (2003). High pressure processing applied to cooked sausage manufacture: Physical properties and sensory analysis. *Meat Science*, 65, 1187–1191.
- Mutungi, G., Purslow, P., & Warkup, C. (1995). Structural and mechanical changes in raw and cooked single porcine muscle fibres extended to fracture. *Meat Science*, 40, 217–234.
- Rust, R., & Olson, D. (1988). Making good "lite" sausage. *Meat and Poultry*, 34, 10–16.
- Ruusunen, M., & Puolanne, E. (2005). Reducing sodium intake from meat products. *Meat Science*, 70, 531–541.
- Scott-Blair, G. W. (1958). Rheology in food research. *Advances in Food Research*, 8, 1–61.
- Suzuki, T. (1981). Kamaboko (fish cake). In *Fish and krill protein. Processing technology* (pp. 62–191). London: Applied Science Publishers Ltd.
- Willems, M. E. T., & Purslow, P. P. (1996). Effect of postrigor sarcomere length on mechanical and structural characteristics of raw and heat-denatured single porcine muscle fibres. *Journal of Texture Studies*, 27, 217–233.
- Yılmaz, I., Simsek, O., & Isikli, M. (2002). Fatty acid composition and quality characteristics of low-fat cooked sausages made with beef and chicken meat, tomato juice and sunflower oil. *Meat Science*, 62, 253–258.